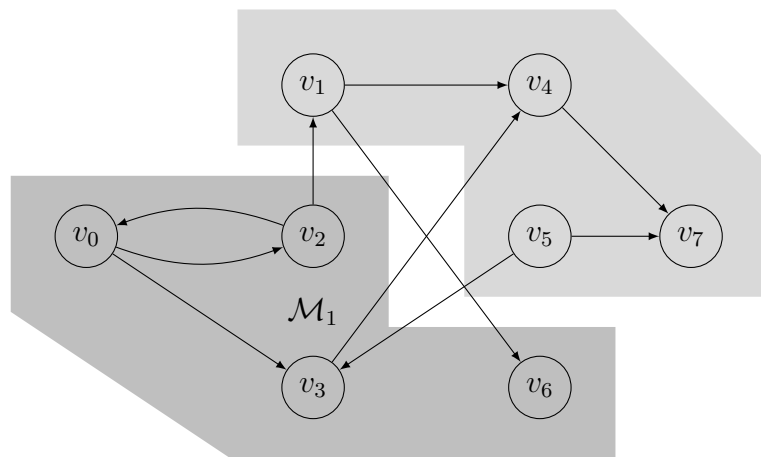


Cuts

Michel Bierlaire

Solution of the practice quiz

1. The first cut Γ_1 is shown in the figure below.



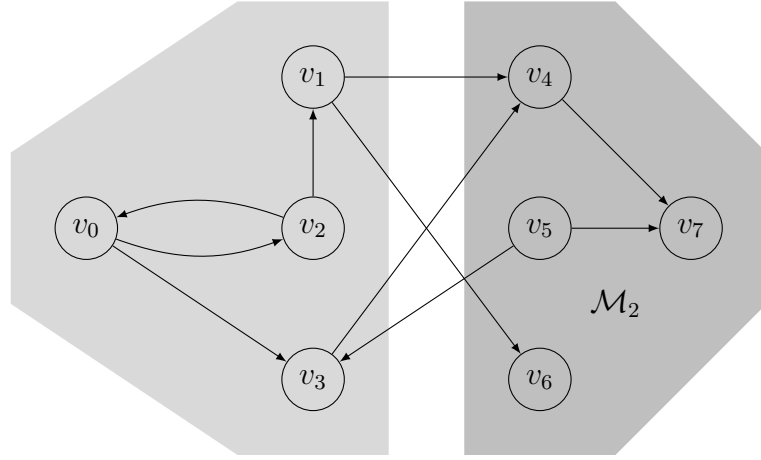
The set of forward arcs is

$$\Gamma_1^{\rightarrow} = \{(v_2, v_1), (v_3, v_4)\},$$

and the set of backward arcs is

$$\Gamma_1^{\leftarrow} = \{(v_1, v_6), (v_5, v_3)\}.$$

2. The second cut Γ_2 is shown in the figure below.



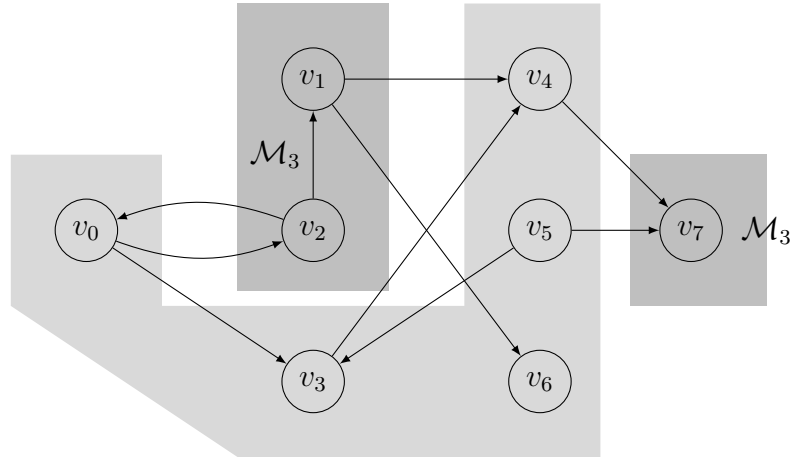
The set of forward arcs is

$$\Gamma_2^{\rightarrow} = \{(v_5, v_3)\},$$

and the set of backward arcs is

$$\Gamma_2^{\leftarrow} = \{(v_1, v_4), (v_1, v_6), (v_3, v_4)\}.$$

3. The third cut Γ_3 is shown in the figure below.



The set of forward arcs is

$$\Gamma_3^{\rightarrow} = \{(v_1, v_4), (v_1, v_6), (v_2, v_0)\},$$

and the set of backward arcs is

$$\Gamma_3^{\leftarrow} = \{(v_0, v_2), (v_5, v_7), (v_4, v_7)\}.$$